

Practical 4 - Cox regression

Survival Analysis

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Datasets required: `pbcbase`, `alloauto`

R packages required: `survival`, `mexhaz`

```
library(survival)
library(mexhaz)
```

Introduction

This practical is in two parts.

Part A. We will fit Cox models to the PBC data used in Practicals 1 and 2, assess the proportional hazards assumption, and estimate survival curves for different values of the covariates.

Part B. We will use a new dataset, called `alloauto`, to investigate the proportional hazards assumption of the Cox model.

Aims

The aims of this session are:

- Fit Cox proportional hazards models and interpret the results.
- Obtain estimates of hazard ratios to compare hazards for individuals with different values of covariates.
- Obtain estimates of survival curves from a Cox regression model for particular values of the covariates.
- Use methods to assess the proportional hazards assumption.

Part A: Primary Biliary Cirrhosis (PBC) data

We will start by estimating the association between treatment group (`treat`) and the hazard for mortality, using Cox regression. Later in the practical we will also use the `bil0` variable. The time origin is the time of randomisation. The variables of interest for this practical are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>datein</code>	Date person entered the study
<code>dateout</code>	Date of the end of follow-up due to either death or censoring
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years
<code>treat</code>	Treatment: 1=placebo, 2=active
<code>bil0</code>	Serum bilirubin (mg/dl), measured at the start of the trial.

Load the data using:

```
pbcb<-read.table("datasets/pbcbase.csv",sep=";",header=T)
```

Question 1

- Write down the form of the hazard assuming a Cox proportional hazards model for a single covariate X (representing treatment).
- Write down the partial likelihood that can be used to estimate the parameter of this model.

In Cox regression, there is a contribution to the partial likelihood from each event time. In this question we will derive the contribution to the partial likelihood at time $t = 0.052$ in the PBC data.

- What is the value of `treat` for the individual who has the event at time $t = 0.052$?
- How many individuals are at risk at time $t = 0.052$?
- What values of `treat` do these individuals have?
- Using the above information find the contribution to the partial likelihood at time $t = 0.052$.

The output from the following will help:

```

pbc.km.treat <- survfit(Surv(time,d)~treat,data=pbc)

summary(pbc.km.treat,censored = T,times = 0.052)

```

Call: survfit(formula = Surv(time, d) ~ treat, data = pbc)

	time	treat=1 n.risk	n.event	survival	std.err	lower 95% CI
	0.052	89.000	1.000	0.989	0.011	0.967
upper 95% CI	1.000					

	time	treat=2 n.risk	n.event	survival	std.err	lower 95% CI
	0.0520	92.0000	1.0000	0.9892	0.0107	0.9685
upper 95% CI	1.0000					

Question 2

Fit the Cox model including treatment as the covariate using the following code, and interpret the results:

```

pbc.cox<-coxph(Surv(time,d)~as.factor(treat),data=pbc)
summary(pbc.cox)

```

Question 3

- Obtain the estimated survival curves in the two treatment groups based on the Cox model using the code below.

```

pbc.survfit=survfit(pbc.cox,newdata=data.frame(treat=c(1,2)))

plot(pbc.survfit,conf.int = T,
     col=c("black","grey"),
     xlab="Time since randomisation (years)", ylab="Survival probability")

legend(x="bottomleft",c("Placebo","Active"),
      col=c("black","grey"),lty=1)

```

- What is the probability of survival beyond time 5 in the two treatment groups? Hint: you can use a summary of `pbcsurvfit`.
- Why do the estimated survival curves have ‘steps’?
- Why do these survival curves differ from the Kaplan-Meier estimates?

Question 4

- Assess the proportional hazards assumption graphically. Hint: you can make use of a non-parametric plot in the same way as you did to assess a Weibull model in Practical 3.
- What do you conclude about the proportional hazards assumption for these data?

Question 5

Next we investigate how the level of bilirubin measured at the start of the trial (`bil0`) is associated with the outcome.

- Write down the form of a hazard model including both treatment group and baseline bilirubin.
- Fit the model and interpret the results.

Question 6

We will now compare the hazards in different types of individual, based on the treatment and baseline bilirubin value.

- What is the hazard ratio comparing: (i) a person in the active treatment group with `bil0=75`, (ii) a person in the active treatment group with `bil0=30`.
- What is the hazard ratio comparing (i) a person in the placebo group with `bil0=75`, (ii) a person in the placebo group with `bil0=30`.
- What is the hazard ratio comparing (i) a person in the active treatment group with `bil0=75`, (ii) a person in the placebo group with `bil0=30`.

Question 7

- Obtain the estimated survival curves for individuals in the two treatment groups with baseline bilirubin value equal to 15, 30 and 75 (these are approximately the 25th, 50th and 75th percentiles). You can do this by extending the code used in question 4.
- What is the estimated probability of survival beyond 5 years for these six example individuals?

Question 8

Fit a Weibull model including treatment and bilirubin levels as covariates. Compare your results from the Cox model with those from a Weibull model.

Part B: Bone marrow transplant data

The `alloauto` dataset is from a study of 101 individuals with advanced acute myelogenous leukemia. 51 of the patients received treatment using their own bone marrow (an autologous bone marrow transplant) and 50 patients received bone marrow from a sibling (an allogenic bone marrow transplant). The event of interest was a composite of death or relapse. The three variables in the dataset are summarised in the table.

Variable	Description
<code>time</code>	Time in months to event or censoring
<code>delta</code>	Event indicator: 0=Censored, 1=Death or relapse
<code>type</code>	Treatment type: 1=allogenic, 2=autologous

Question 1

Load the data. Obtain Kaplan-Meier estimates of the survival curves in the two treatment groups and perform a log rank test. Interpret your results.

Question 2

- Fit a Cox model including treatment type and interpret the results.
- Use graphical methods to investigate whether the proportional hazards assumption is appropriate for treatment type.